

Last name _____

First name _____

LARSON—MATH 310—CLASSROOM WORKSHEET 11
Gram-Schmidt Algorithm.

Review

1. What does it mean for a set of vectors to be *orthogonal*?
2. What does it mean for a set of vectors to be *orthonormal*?
3. Check that $\vec{e}_1, \vec{e}_2, \dots, \vec{e}_n$ are an orthonormal set (in $\mathbb{R}^n$).
4. Why are orthonormal vectors linearly independent?
5. What formula did we learn for the coefficients of a vector $\vec{x}$ with respect to an orthonormal basis $\vec{a}_1, \vec{a}_2, \dots, \vec{a}_n$?

New (Chp. 5)

1. What is the *orthonormal expansion* of $\vec{x} = \begin{bmatrix} 5 \\ -2 \\ 1 \end{bmatrix}$ with respect to the orthonormal list $\vec{e}_1, \vec{e}_2, \vec{e}_3$?

2. Apply the Gram-Schmidt algorithm to the input set $\vec{a}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \vec{a}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

3. Are the (1) input vectors linearly dependent, or (2) the output vectors are orthonormal? Check.

4. Apply the Gram-Schmidt algorithm to the input set $\vec{a}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \vec{a}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \vec{a}_3 = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$.
What do we learn?